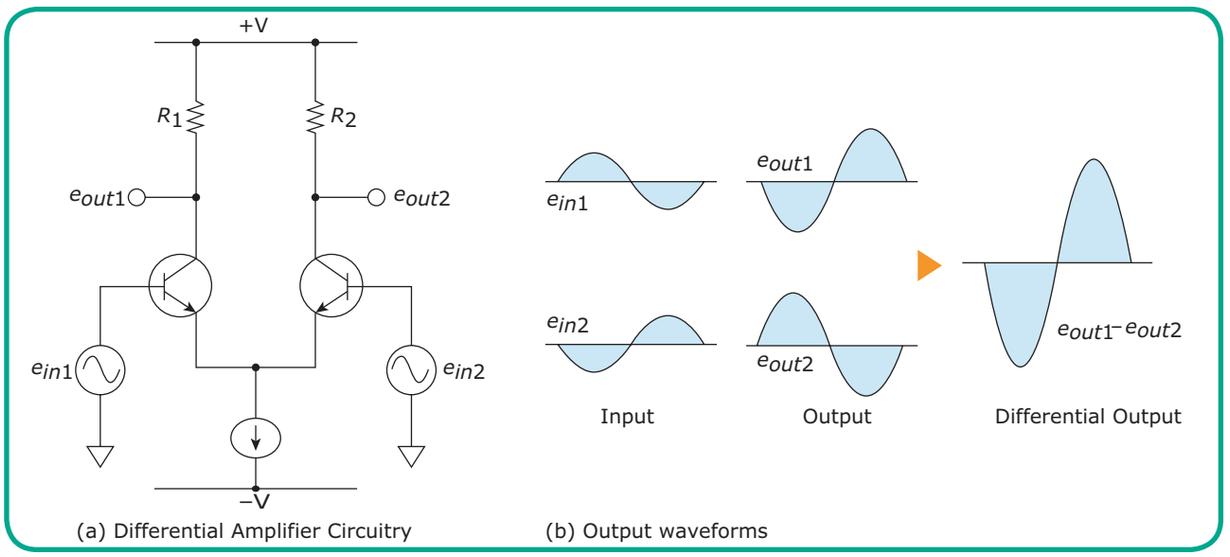


Figure: 5.30 **Differential Amplifier**

5.13 FEED BACK IN AMPLIFIERS

The process of injecting a fraction of output energy back to the input is known as feedback

There are two basics types of feedback in amplifiers

1. Positive Feedback
2. Negative Feedback

5.13.1 Positive Feedback

When the feedback energy (voltage or current) is in phase with the input signal and thus enhances the input signal, the feedback is called positive feedback or regenerative feedback. This is illustrated in Figure 5.31.

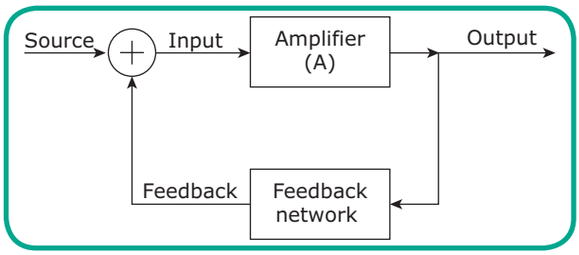


Figure 5.31 **Positive Feedback**

5.13.2 Negative Feedback

When the feedback energy (voltage or current) is out-off phase with the input signal and thus oppose the input signal, the feedback is called negative feedback or degenerative feedback. This is illustrated in Figure 5.32.

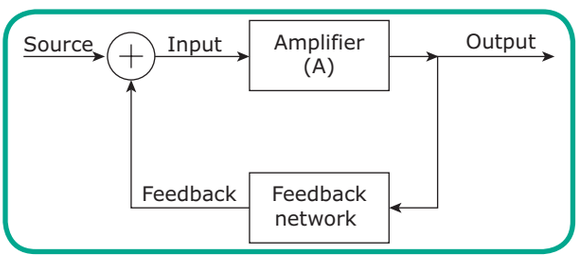


Figure 5.32 **Negative Feedback**

5.14 DISTORTION IN AMPLIFIERS

The change in output wave-shape from the input wave-shape of an amplifier is known as distortion.

The distortion can be classified as follows

1. Amplitude distortion
2. Phase distortion
3. Frequency distortion

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